



2025 State of the API

for Security



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246 security professionals weigh in on the state of APIs in the age of AI

For decades, security teams have navigated a complex balancing act: protecting organizations from ever-evolving threats while enabling the business to move forward. Despite this, our recent survey suggests that security teams are no longer cautious gatekeepers in fast-paced development environments; they're the trailblazers of innovation.

We surveyed 246 security professionals who are redefining their role in the API ecosystem. These aren't security teams focused solely on defense and risk mitigation. They're strategic architects constructing secure foundations for the AI revolution.

Our data reveals something unexpected: In the age of AI and API-first development, security teams aren't just participants; they're pioneers, outpacing other functions in critical areas that drive both protection and business value.

The traditional view of security as primarily protective is expanding. Today's security professionals are becoming catalysts for secure innovation, particularly at the intersection of APIs and AI.



Security teams lead across three pivotal dimensions

Our analysis identifies three areas where security professionals are establishing decisive momentum, often surprising their peers in development, product, and IT:

1. AI agent readiness is no longer optional; it's a security imperative

73% of security professionals evaluate their APIs for AI agent compatibility, putting them 28 percentage points ahead of the overall average and 41 percentage points ahead of their engineering and software development counterparts—the largest differential in our entire survey. While other roles debate whether AI agents will matter, security teams are already building protective guardrails for this inevitable future.

2. Security leaders report greater revenue impact from APIs, positioning security as a business enabler

Security professionals report significantly higher API revenue impact than their peers. Eighty-seven percent say APIs generate revenue for their organization (vs. 70% average), with 30% reporting that APIs generate 51-75% of total revenue (vs. just 13% overall). Security is evolving beyond risk management to become a value enabler that protects and enhances revenue-generating assets.

3. MCP adoption proves security teams are building for the AI future

31% of security professionals use Model Context Protocol (MCP) regularly, more than double the 14% average across all disciplines. This signals that security teams understand the need for a secure intermediary layer between AI agents and APIs, and are implementing solutions ahead of the curve.

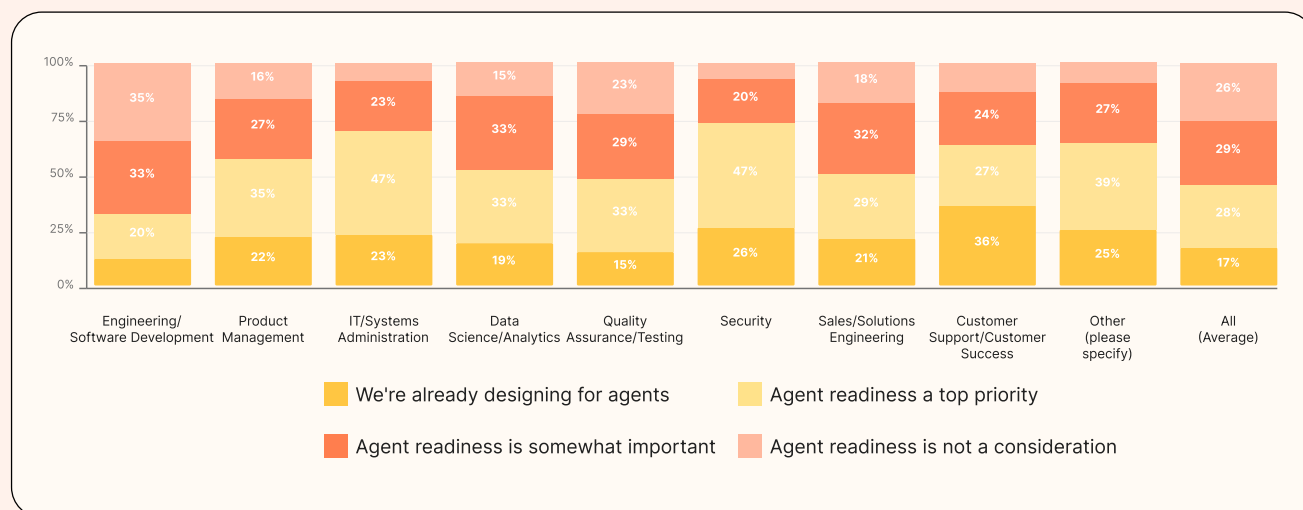
The question facing organizations isn't whether security should have a seat at the API strategy table—it's whether they recognize that security teams are already reshaping how APIs and AI interact, often leading the conversation rather than following it.

Let's examine what your peers shared in our survey.



Security professionals are building for AI agents, right now

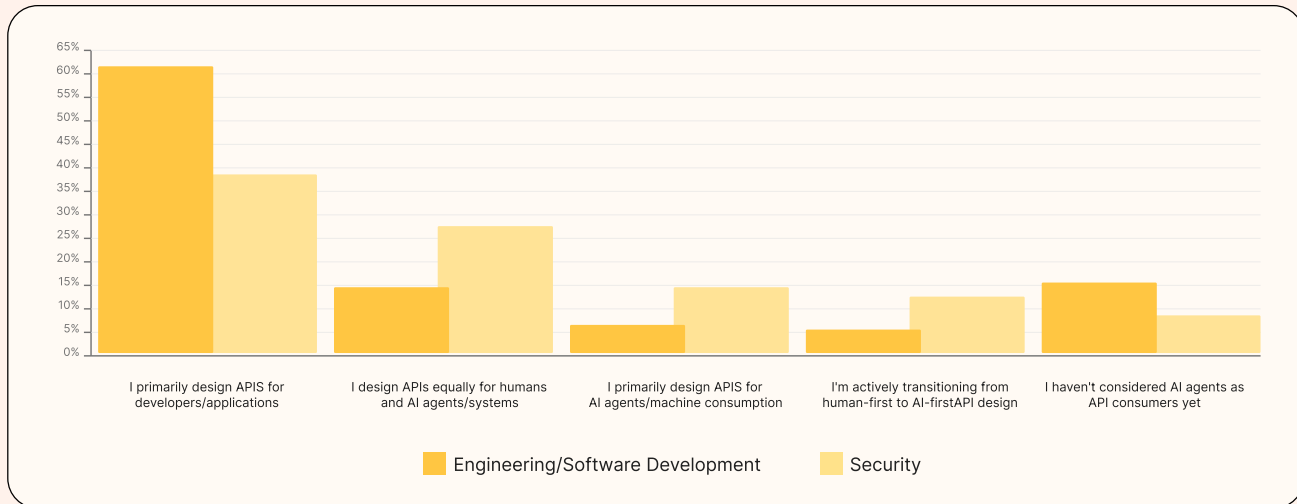
The data shows that 73% of security professionals are actively preparing for AI agents (26% already designing for agents, 47% making it a top priority), compared to just 45% across all other disciplines combined. That's not just a modest difference; it's a different mindset entirely.



When we specifically compare security professionals to engineering/software development teams—traditionally considered the early adopters of new technologies—the contrast becomes even more striking:

- **Already designing for agents:** 26% security vs. only 12% of engineering/development teams
- **Agent readiness as top priority:** 47% security vs. just 20% of engineering/development teams
- **Agent readiness NOT a consideration:** Only 7% security vs. 35% of engineering/development teams

This 27-point gap between security and engineering teams on prioritizing agent readiness represents one of the largest functional divides in our survey. While developers focus primarily on human-facing APIs (61% design primarily for humans vs. only 38% of security professionals), security teams are preparing for a future where machine consumers dominate.



While it's promising to see that roughly a quarter of security teams are already designing for AI agents, there's still room for growth. The most forward-thinking security organizations are paving the way, creating AI-ready security models that protect both human and non-human API consumers.

How Postman helps protect your APIs from human and non-human consumers

In order to pull ahead, Postman provides security-minded professionals the tools they need to confidently protect their APIs:

- **Security built in, not bolted on:** Teams using Postman can design, test, and secure APIs before they're deployed, preventing risk long before runtime. Security isn't limited to gateway or perimeter tools; it's enforced throughout the lifecycle:
 - With **Spec Hub**, you can enforce schema validation, versioning, and naming conventions—catching inconsistencies early.
 - Postman **API Security** automatically scans for vulnerabilities aligned with OWASP Top 10 for APIs (auth leaks, injection, data exposure, etc.).
 - **Secret Scanner** and **Postman Local Vault** prevent credential exposure and enforce least privilege access by design.
- **AI-ready authentication and access control:** AI agents call APIs at machine speed and scale, so Postman enables identity management for both humans and machines:
 - Environment-scoped secrets, tokens, and RBAC ensure AI agents access only the APIs they're authorized to invoke.
 - MCP servers act as secure intermediaries between APIs and AI agents, exposing only whitelisted endpoints and data, limiting scope and surface area. (More on this later.)
 - Dynamic rate-limiting and behavioral observability (through **Insights** and **Monitors**) detect non-human call patterns or anomalies.

With Postman, security teams can assume caution without risk and be at the forefront of preparing API infrastructure for the next generation of consumers.

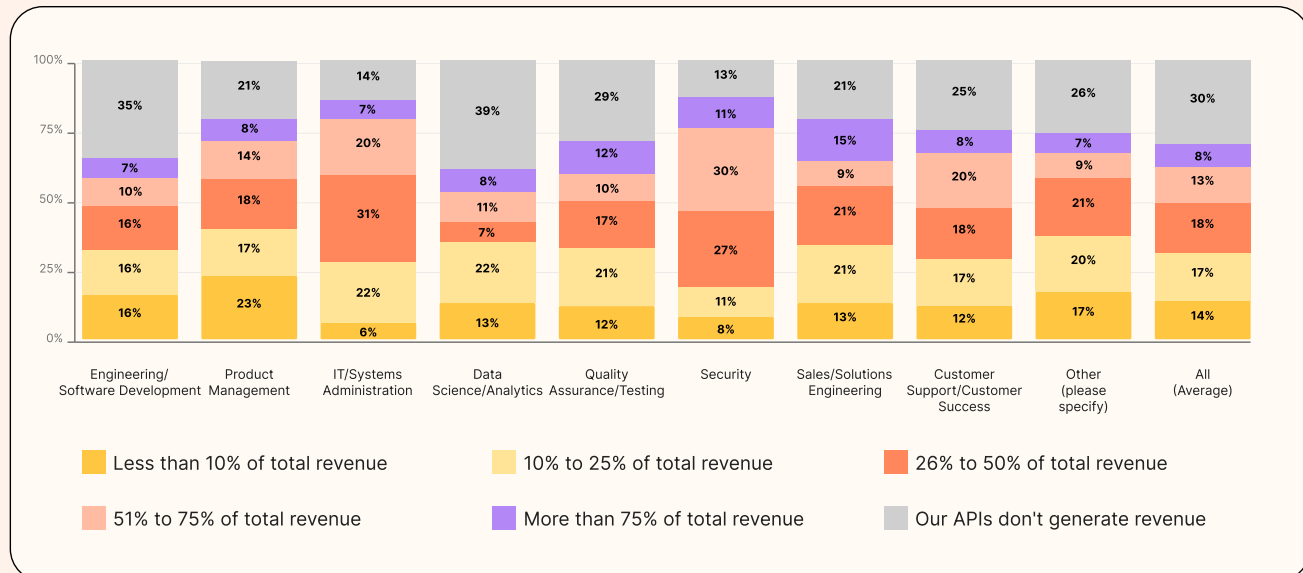


APIs are revenue engines, even in security's view

According to the [2025 State of the API report](#), 65% of organizations now generate revenue from their APIs, so there's no argument about their monetary value. However, security teams traditionally viewed APIs primarily as potential attack vectors. But our data reveals a dramatic shift in perspective: Security professionals view APIs as strategic business assets that drive significant revenue.

According to our survey, security professionals report APIs generating substantial revenue. The revenue impact breakdown shows security professionals recognize APIs' financial contribution:

- 27% say APIs generate 26-50% of revenue (vs. 18% average across all job functions)
- 30% say APIs generate 51-75% of revenue (vs. 13% average across all job functions)
- 13% say APIs don't generate revenue (vs. 30% average across all job functions)





APIs generate revenue

This recognition of APIs as revenue drivers appears to influence investment decisions. Sixty-two percent of security professionals report their organizations will increase API investments in the next 12 months, compared to 49% across the average of all other job functions surveyed—a clear sign that protecting and enhancing these assets is now a business imperative, not just a technical requirement.



Case Study

For Youi, an insurance company, Postman provides their various application teams with a secure and effective way of collaborating on APIs in an always-up-to-date environment. Simultaneously, experts in compliance regulations can provide instant feedback to ensure API adherence while the Architecture teams create Open API specifications. With Postman, it takes three or four clicks to import specific and relevant APIs, whereas previously, it took days.

[Read the full story →](#)

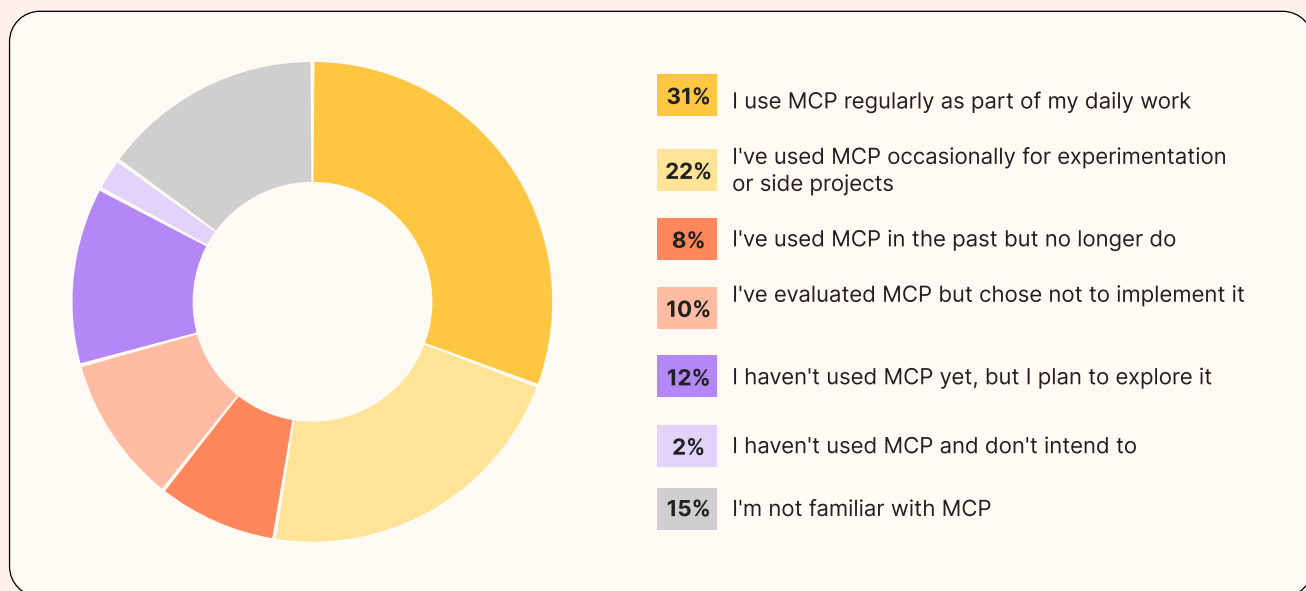


Security professionals lead Model Context Protocol (MCP) adoption

MCP is emerging as the critical connective layer between AI agents and APIs, allowing machines to discover, understand, and safely invoke APIs. According to the 2025 State of the API report, while 70% of developers are aware of MCP, only 10% are using it regularly.

However, security professionals stand out as early adopters:

- 31% use MCP regularly (vs. 14% average across all job functions)
- 22% have used MCP occasionally (in line with the average of all job functions)
- Only 15% are unfamiliar with MCP (vs. 26% average across all job functions)



This leadership in MCP adoption suggests security teams recognize its value as a protective layer that can control how AI agents interact with APIs, enforcing governance policies programmatically rather than relying solely on traditional security measures.



Why Postman enables MCP

Think of MCP as a firewall with context for AI. Instead of an LLM directly touching your API surface, it can only act through an MCP server that mediates every request.

Without MCP, an LLM or agent might “scrape” or brute-force APIs blindly.

With MCP, you define:

- Which endpoints are exposed
- What authentication they require
- What input/output schemas exist
- What policies or limits apply



Did you know?

15k MCP servers have been generated using Public Postman Collections. Try our no-hassle way of using existing API documentation on Postman to create MCPs.

[Learn more →](#)

As security professionals lead the adoption of technologies like MCP and the evaluation of APIs for AI compatibility, a profound mindset shift is occurring throughout the industry.



Security teams are powering the AI-API future

The data is clear: Security professionals aren't just participants in the AI-API revolution—they're leading it in critical dimensions that bridge protection and innovation.

Security teams today are developing new models that accommodate both human and AI consumers, recognizing that tomorrow's threats and opportunities will come from both.

The data shows security professionals are particularly focused on controlling how AI agents interact with APIs—enforcing boundaries that protect sensitive data while enabling innovation.

Three actions for forward-thinking security leaders:

1. **Reposition security as a business enabler:** Frame API security as protecting revenue streams, not just preventing breaches. When 87% of security professionals recognize APIs generate revenue, they're acknowledging security's role in business value creation.
2. **Lead the AI agent readiness initiative:** Don't wait for development teams to consider agent compatibility. Security's 27-point lead over engineering in evaluating APIs for agent compatibility shows security can drive this conversation.
3. **Invest in MCP:** With 31% of security professionals already using MCP regularly, security teams are establishing expertise in this critical AI-API intermediary layer. This knowledge will become increasingly valuable as AI adoption accelerates.

The most successful organizations will be those that leverage their security teams' foresight to build API ecosystems that are not just secure by design, but ready for the AI-powered future.

